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GCSE · OCR · Computer Science

2 hours 32 questions

Programming Fundamentals & Data Types

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Fundamental Programming Concepts / Programming Constructs / Common Arithmetic Operators / Common Boolean Operators / Data Types

Easy (11 questions)	/34
Medium (14 questions)	/32
Hard (7 questions)	/28
Total Marks	**/94**

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Easy Questions

1 **Tick (✓) one box in each row** to identify whether the OCR Reference Language code given is an example of selection or iteration.

OCR Reference Language code	Selection	Iteration
for i = 1 to 10 print(i) next i		
while score != 0 playgame() endwhile		
if playerHit() then score = 0 endif		
switch bonus: case 0: score = 9 case 1: score = 7 case 2: score = 5 endswitch		

(4 marks)

2 Jack's program uses the addition (**+**) **arithmetic operator**.

This adds together two numbers.

State the purpose of each of the following **arithmetic operators**

- *
- /

.....

..... **(2 marks)**

3 (a) Customers at a hotel can stay between 1 and 5 (inclusive) nights and can choose between a basic room or a premium room.

A typical booking record is shown in the table:

firstName	Amaya
surname	Taylor-Ling
nights	3
room	Premium
stayComplete	False

State the most **appropriate data type** for the following fields:

Nights	
Room	

.....

..... **(2 marks)**

(b) **Give** the name of **one field** that could be stored as a **Boolean data type**.

..... **(1 mark)**

4 (a) A program is written to allow a user to enter the radius of a circle as a whole number between 1 and 30, then calculate and output the area of the circle.

Identify two variables used in the program

```
radius = 0
area = 0.0
radius = input("Enter radius")
if radius < 1 OR radius > 30 then
print("Sorry, that radius is invalid")
else
area = 3.142 * (radius ^ 2)
print (area)
endif
```

.....

..... **(2 marks)**

(b) **Identify one item** in the program that could have been written as a **constant**.

..... **(1 mark)**

(c) **Give one reason** why you have identified this item as a constant.

..... **(1 mark)**

(d) **Tick (✓) one** box **in each row** to identify whether each **programming construct** has or has not been used in the program in **part (a)**.

	Has been used	Has not been used
Sequence		
Selection		
Iteration		

(3 marks)

5 A teacher asks students how long they spend completing homework. Students answer in minutes and hours (for example 2 hours 15 minutes).

The teacher would like to create an algorithm that will display students' inputs in minutes only.

Identify the **input** and **output** required from this algorithm.

.....

..... **(2 marks)**

6 (a) **State what is meant** by a **real data type** and **give an example** of this data type.

.....

..... **(2 marks)**

(b) **State what is meant** by an **integer data type** and **give an example** of this data type.

.....

..... **(2 marks)**

7 A programmer declares the following variables.

first = "Computer Science"

second = "is great"

State one difference between a **variable** and a **constant**.

..... **(1 mark)**

8 (a) **Write a pseudocode** statement to **assign the value** 7.3 to a **variable** with the **identifier** timer

..... **(1 mark)**

(b) **State** the **most appropriate data type** for the variable timer.

..... **(1 mark)**

9 OCR High School uses a computer system to store data about students' conduct.

The system records good conduct as a positive number and poor conduct as a negative number.

A TRUE or FALSE value is also used to record whether or not a letter has been sent home about each incident.

An example of the data held in this system is shown below in **Fig. 1**:

StudentName	Detail	Points	LetterSent
Kirstie	Homework forgotten	−2	FALSE
Byron	Good effort in class	1	TRUE
Grahame	100% in a test	2	FALSE
Marian	Bullying	-3	TRUE

State the **most appropriate data type** used to store each of the following items of data.

StudentName	
Points	
LetterSent	

.....

.....

..... **(3 marks)**

10 **Identify two** basic **programming constructs** that have been used in this algorithm.

```
for k = 1 to 3
for p = 1 to 5
print (k + p)
next p
```

```
05 next k

06 m = 7

07 print m * m
```

1	
2	

.....

..... **(2 marks)**

11 An infinite loop is where a section of a program repeats indefinitely.

For each of the pseudocode algorithms shown below, **tick the appropriate box** to show whether they will loop infinitely or not.

Pseudocode	Will loop infinitely	Will not loop infinitely
01 x = 0 02 while True 03 print x 04 endwhile		
01 x = 0 02 while x < 10 03 print x 04 endwhile		
01 x = 0 02 while x < 10 03 print x 04 x = x + 1 05 endwhile		
01 y = 5 02 for x = 1 to y 03 print x 04 next		

(4 marks)

Medium Questions

1 A teacher researches the length of time students spend playing computer games each day.

Tick (✓) one box to identify the **data type** you would choose to **store the data** and **explain why** this is a suitable data type.

Data Type	Tick (✓) one box
String	
Integer	
Real	
Boolean	

Explanation:	

.....

..... **(2 marks)**

2 (a) DIV and MOD are both operators used in computing-related mathematics.

State the **value** of 13 DIV 4

..... **(1 mark)**

(b) **State** the **value** of 13 MOD 4

.....

(1 mark)

3 The symbol **^** is used for **exponentiation**.

Give the result of a^b when a = 3 and b = 2.

.....

(1 mark)

4 Each member of staff that works in a restaurant is given a Staff ID. This is calculated using the following algorithm.

```
surname = input("Enter surname")
year = input("Enter starting year")
staffID = surname + str(year)
while staffID.length < 10
    staffID = staffID + "x"
endwhile
print("ID " + staffID)
```

Define the term **casting** and **give the line number** where casting has been used in the algorithm.

.....

.....

(2 marks)

5 **Tick (✓)** one box in each row to identify the programming construct where each keyword is used.

Keyword	Programming construct	
	Selection	Iteration
if		
for		
while		

(3 marks)

6 The students are put into teams.

Students gain points depending on their result and the year group they are in.

The points are added to the team score.

The team with the most points at the end of the sports day wins.

Data about the teams and students is stored in a sports day program.

Identify the most appropriate data type for each variable used by the program.
Each data type must be different.

Variable	Example	Data type
teamName	"Super-Team"	
studentYearGroup	11	
javelinThrow	18.2	

(3 marks)

7 A library system assigns unique Member IDs to new users. The following algorithm is used to generate these IDs.

```
01  firstname = input("Enter first name")
02  month = input("Enter birth month as number")
03  memberID = firstname + str(month)
04  while memberID.length < 12
05      memberID = memberID + "z"
06  endwhile
07  print("Member ID: " + memberID)
```

Define the term **casting** and **give the line number** where casting has been used in the algorithm.

.....

..... **(2 marks)**

8 A library management system assigns unique Member IDs to new library users. The following algorithm is used to generate these IDs:

```
01  firstName = input("Enter first name")
02  lastName = input("Enter last name")
03  membershipYear = input("Enter membership year")
04  tempID = firstName[0] + lastName + str(membershipYear)
05  memberID = tempID.upper()
06  counter = 1
07  while len(memberID) < 12
08      memberID = memberID + str(counter)
09      counter = counter + 1
10  endwhile
11  finalID = memberID[:12]
12  print("Member ID: " + finalID)
```

Define the term **casting** and **identify all line numbers** where casting has been used in this algorithm. **Explain** why casting is necessary in each instance you have identified.

.....

.....

.....

.....

..... **(5 marks)**

9 **Describe** one reason why a programmer would use a constant instead of a variable for the value of pi (π)

.....

..... **(2 marks)**

10 **Explain** the importance of using a string prompt inside an input() statement

.....

..... **(2 marks)**

11
```
print(total)
total = total + price
```
Describe how the order of these lines affects the logic of the algorithm

.....

..... **(2 marks)**

12
```
if mark ≥ 70 then
   grade = "A"
elseif mark ≥ 60 then
   grade = "B"
endif
```
Describe the role of the elseif statement in this code

.....

..... **(2 marks)**

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13 Consider the two loops below:

```
for i = 1 TO 3
   password = input("Enter password")
next i

while password != "secure"
   password ← input("Enter password")
endwhile
```

Explain why a while loop is more suitable than a for loop in this situation

.....

..... **(2 marks)**

14 **Identify** the result of the Boolean expression

(X > 5) AND (Y < 10)

if X = 7 and Y = 12

.....

..... **(2 marks)**

Hard Questions

1 **Identify** and **correct** the errors in the **Python** program below.

Explain why each correction is necessary.

```
NAME = "Alice"

PI = 3.14159

name = "Bob"

NAME = "Charlie"

radius = 10

area = PI * radius ^ 2

print("The area of the circle is", area)
```

.....

.....

.....

..... **(4 marks)**

2 **Modify** the Python code to include **a selection construct** that **only adds even numbers** to the total.

```
total = 0

for i in range(5):

num = int(input("Enter a number: "))

total = total + num

print("Total is:", total)
```

.....

..... **(2 marks)**

3 (a) **Identify** the **output** of each print statement in the following Python program:

```
a = True

b = False

print(a and b)

print(a or b)

print(not a)
```

Statement	**Output**
print(a and b)	
print(a or b)	
print(not a)	

(3 marks)

(b) **Modify** the **code** in part(a) to include a conditional that prints:

"Both are True" if both a and b are true

AND

"One is True" if either a or b is true but not both

.....

.....

.....

..... **(4 marks)**

4 Students take part in a sports day. The students are put into teams.

Students gain points depending on their result and the year group they are in. The points are added to the team score.

The team with the most points at the end of the sports day wins.

An algorithm works out which team has won (has the highest score).

Write an algorithm to:

- Prompt the user to enter a team name and score, or to enter **"stop"** to stop entering new teams
- Repeatedly take team names and scores as input until the user enters **"stop"**
- Calculate which team has the highest score
- Output the team name and score of the winning team in an appropriate message.

You must use **either**:

- OCR Exam Reference Language, **or**
- A high-level programming language that you have studied

.....

.....

.....

.....

.....

..... **(6 marks)**

5
```
balance = balance − amount
if balance ≥ 0 then
   print("Withdrawal approved")
endif
```
Explain the possible impact of this sequence error

.....

.....

..... **(3 marks)**

6
```
do
   password = input("Enter password")
until password == "secure"
```
Evaluate why a do until loop may be more suitable than a while loop for this system

.....

.....

..... **(3 marks)**

7 **Justify** why the Real (Float) data type should be used instead of Integer to store the result of a division

.....

.....

..... **(3 marks)**

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